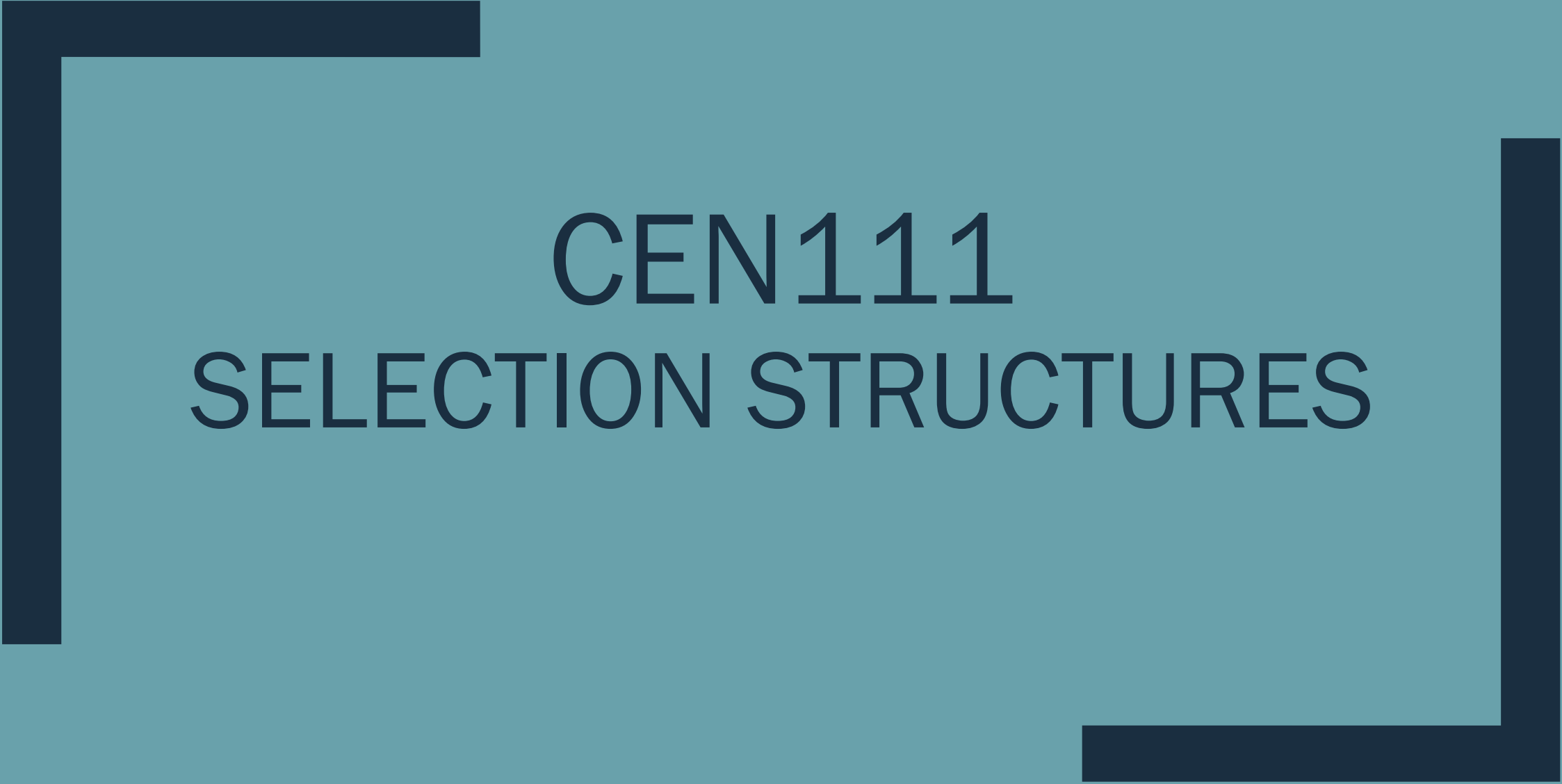


The image features a light blue background with two dark blue L-shaped brackets. One bracket is in the top-left corner, and the other is in the bottom-right corner. They are positioned such that they appear to frame the central text.

CEN111

SELECTION STRUCTURES

Structures

- Sequence structures:
 - *Programs executed sequentially by default*
- Selection structures:
 - *if, if...else, and switch*
- Repetition structures:
 - *while, do...while and for*

Control Structures

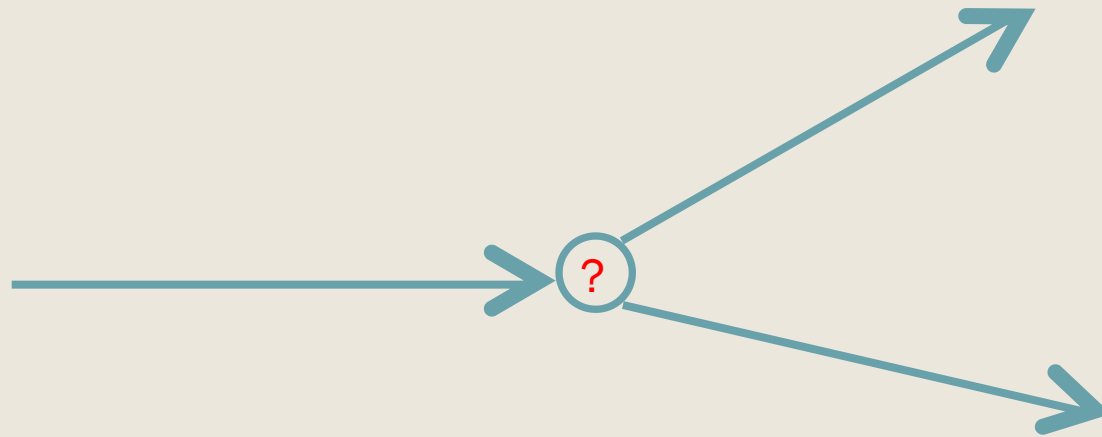
- control structure
- compound statement
 - *a group of statements bracketed by { and } that are executed sequentially*

Compound Statement

```
{  
    statement;  
    statement;  
    .  
    .  
    .  
    statement;  
}
```

Control Structures

- selection control structure
 - *a control structure that chooses among alternative program statements*



Conditions

- an expression that is either false
 - *represented by 0*
- or true
 - *usually represented by 1*

rest_heart_rate > 75

Relational and Equality Operators

Operator	Meaning	Type
<	less than	relational
>	greater than	relational
<=	less than or equal to	relational
>=	greater than or equal to	relational
==	equal to	equality
!=	not equal to	equality

Relational Operators

- The relational operators are $<$, $>$, $<=$, and $>=$.
- Take 2 expressions as operands
- Yield either the int value 0 (false) or the int value 1 (true).

Example: assume $a = 1$, $b=2$.

Expression	Value
$a <= b$	1
$a < b-5$	0
$a + 10 / b <= -3 + 8$	0

Equality Operators

- The equality operators are == and !=.
- Yield either the int value 0 or the int value 1.

Examples: assume a=1, b=2, ch = 'A'

Expression	Value
a == b	0
a != b	1
ch < 'B'	1
a+b == -2 * 3	0

Equality Operators

- Note carefully that the two expressions **a == b** and **a = b** are visually similar.
- The expressions
 - *a == b is a test for equality.*
 - *a = b is an assignment expression*

Logical Operators

- logical expressions
 - *an expression that uses one or more of the logical operators*
 - && (and)
 - || (or)
 - ! (not)

Logical Operators

- Expressions connected by && or || are evaluated left to right.
- logical complement (negation):!
 - *the complement of a condition has the value 1 (true) when the condition's value is 0 (false)*
 - *the complement of a condition has the value 0 (false) when the condition's value is nonzero (true)*

Logical Operators

Expression	Value
!5	0
!!5	1
!(6 < 7)	0
!6 < 7	1
!(3-4)	0

Operator Precedence

Operator	Precedence
()	highest (evaluated first)
! + - (unary operator)	
* / %	
+ -	
< <= >= >	
== !=	
&&	
=	lowest (evaluated last)

int i=5, j=10, k=20;

$i > 5 * k \% 3 \ \&\& \ !(\underline{k-2} != 18)$

↓

$i > 5 * k \% 3 \ \&\& \ !(\underline{18} != 18)$

↓

$i > 5 * k \% 3 \ \&\& \ !\underline{0}$

↓

$i > \underline{5 * k} \% 3 \ \&\& \ 1$

↓

$i > \underline{100 \% 3} \ \&\& \ 1$

↓

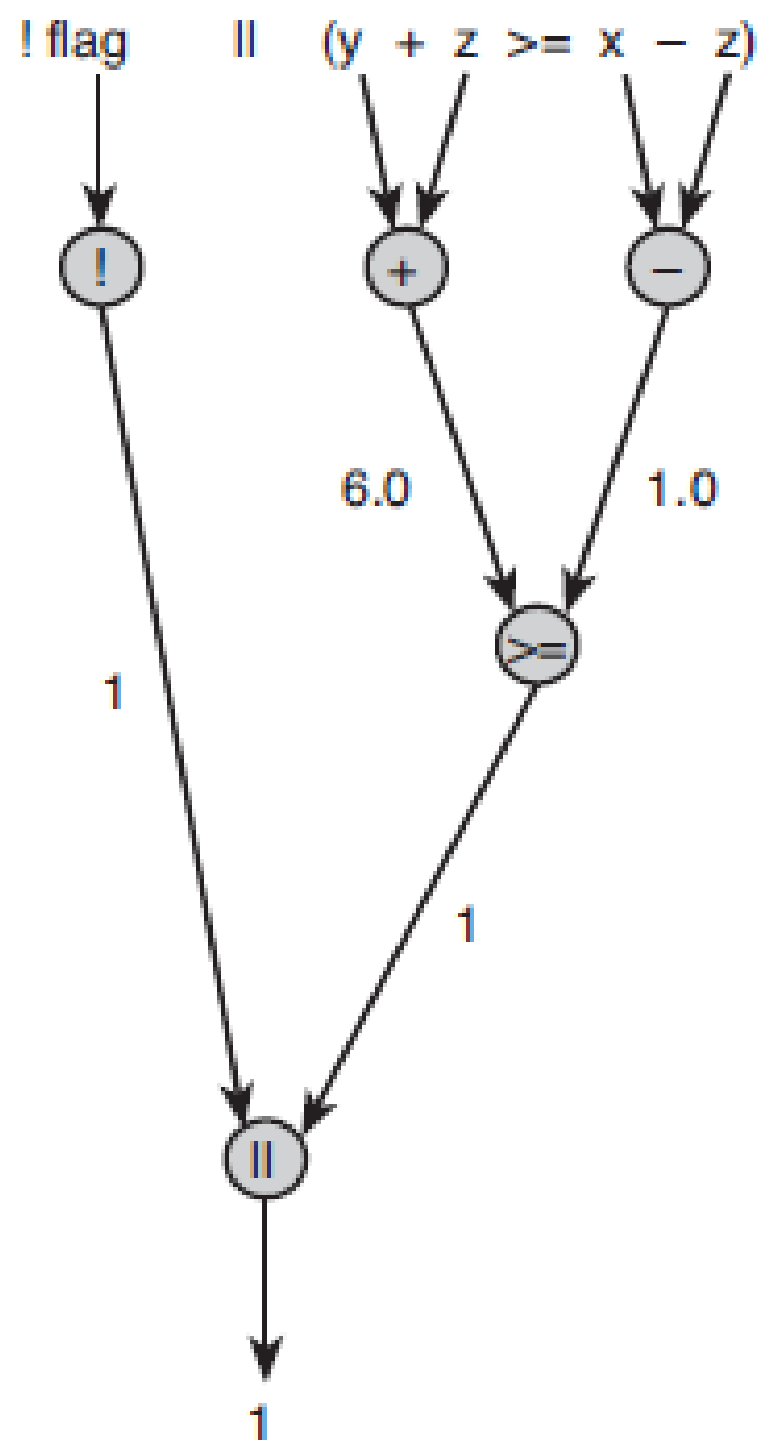
$\underline{i > 1} \ \&\& \ 1$

↓

$\underline{1 \ \&\& \ 1}$

↓

1



flag	y	z	x
0	4.0	2.0	3.0

!flag		(y + z	>=	x - z)
<u>0</u>		<u>4.0 2.0</u>		<u>3.0 2.0</u>
1		<u>6.0</u>		<u>1.0</u>
<u>1</u>				

Short-Circuit Evaluation

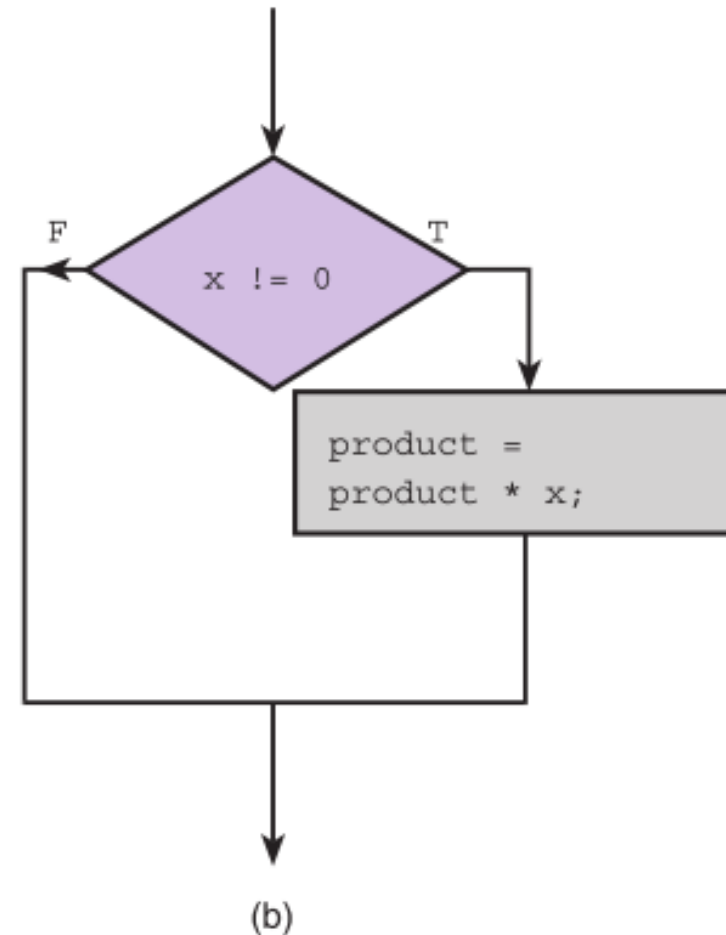
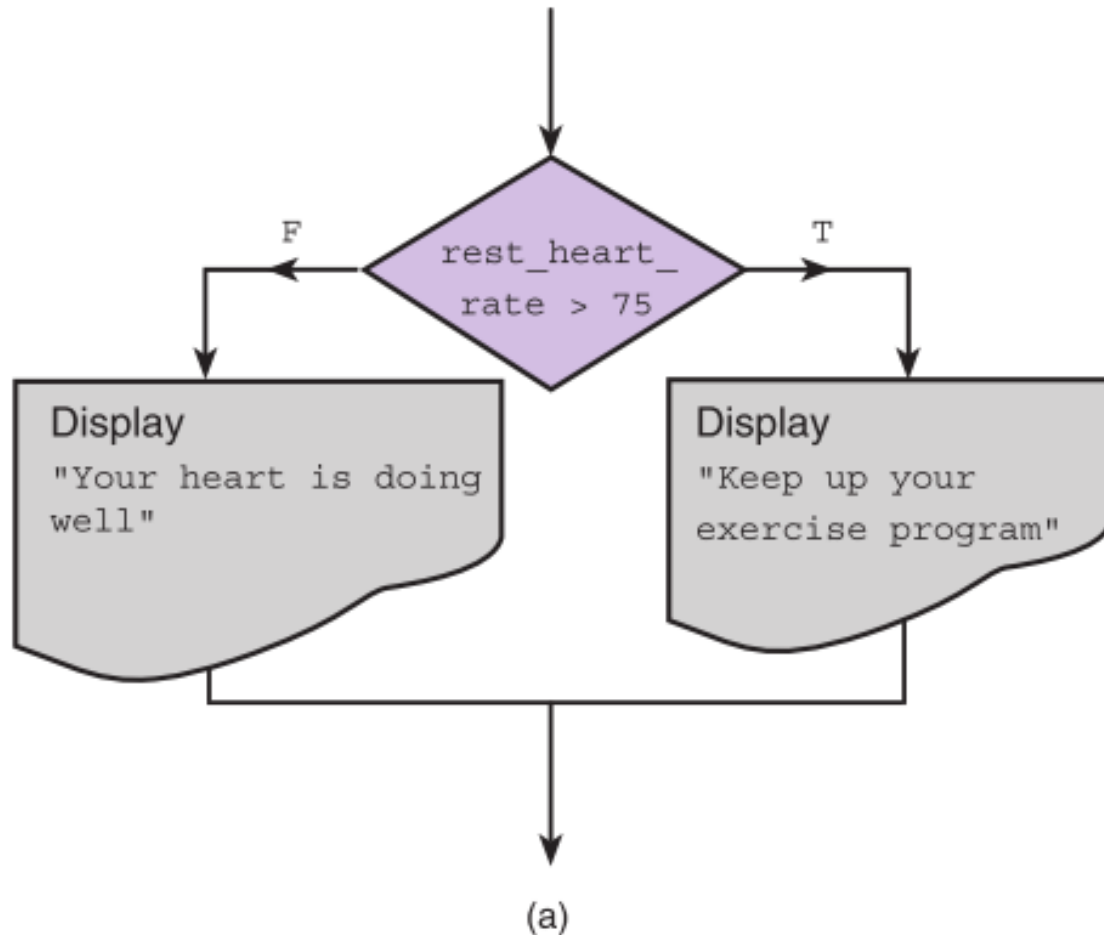
- For the expressions that contain the operands of `&&` and `||`, the expression process stops as soon as the outcome true or false is known.
- Suppose `expr1` is 0.
 - `expr1 && expr2 = 0` (*expr2 will not be evaluated.*)
- Suppose `expr1` is nonzero.
 - `expr1 || expr2 = 1` (*expr2 will not be evaluated.*)

SELECTION STRUCTURE

making decisions



Flowcharts of if Statements with Two Alternatives and One Alternative



The if Selection Statement

FORM: if (condition)

statement_T ;

EXAMPLE: if (x > 0)

prod = prod * x;

INTERPRETATION: If condition evaluates to true (a nonzero value), then statement_T is executed; otherwise, statement_T is skipped.

The if...else Selection Statement

FORM: if (condition)
 statement_T ;
 else
 statement_F ;

EXAMPLE: if (x > 0)
 printf("positive\n");
 else
 printf("negative\n");

INTERPRETATION: If condition evaluates to true (a nonzero value), then statement_T is executed and statement_F is skipped; otherwise, statement_T is skipped and statement_F is executed.

The if...else Selection Statement

- if
 - *Only performs an action if the condition is true*
- if...else
 - *Specifies an action to be performed both when the condition is true and when it is false*

The if...else Selection Statement

- Compound statement:
 - *Set of statements within a pair of braces*

- Example:

```
if ( grade >= 60 )
    printf( "Passed.\n" );
else
{
    printf( "Failed.\n" );
    printf( "You must take this course again.\n" );
}
```

The if..else Selection Statement

- Ternary conditional operator (?:)
 - *Takes three arguments (condition, value if true, value if false)*

```
grade >= 60 ? printf( "Passed\n" ) : printf( "Failed\n");
```


Nested if/else structures

- Test for multiple cases by placing if...else selection statements inside if...else selection statement
- Once condition is met, rest of statements skipped
- Deep indentation usually not used in practice

Nested if/else structures

If student's grade is greater than or equal to 90

Print "A"

else

If student's grade is greater than or equal to 80

Print "B"

else

If student's grade is greater than or equal to 70

Print "C"

else

If student's grade is greater than or equal to 60

Print "D"

else

Print "F"

Nested if/else structures

```
if (expr1)
    statement1
else if(expr2)
    statement2
else if(expr3)
    statement3
.....
else if(exprN)
    statementN
else
    default statement
next statement
```

Nested if/else structures

```
if(grade >= 90)
    printf("A");
else if (grade >= 80)
    printf("B");
else if (grade >= 70)
    printf("C");
else if (grade >= 60)
    printf("D");
else
    printf("F");
```

The switch Multiple-Selection Structure

- used to select one of several alternatives
- useful when the selection is based on the value of
 - *a single variable*
 - *or a simple expression*
- values may of type int or char
 - *not double*

The switch Multiple-Selection Structure

```
switch ( a_variable ){  
  case value1:  
    actions;  
    break;  
  case value2 :  
    actions;  
    break;  
  ...  
  default:  
    actions;  
}
```

Exercise

- Write a simple calculator using the switch multiple-selection structure